Problem Solving with Centripetal Acceleration

Answer key

Conventions used throughout

• **Formulas:** $a_c = v^2/r = 4\pi^2 r/T^2 = 4\pi^2 r f^2$; $v = 2\pi r/T$; $\Sigma F_c = ma_c$; universal gravitation $F_g = GMm/r^2$, where r is measured from centre to centre.

• **Orbits:** gravity is the only force on an orbiting object, so it is the centripetal force: $GMm/r^2 = mv^2/r = m4\pi^2 r/T^2$. The orbiting object's mass m always cancels.

• **Constants:** $G = 6.67 \times 10^{-11}$ N·m^2/kg^2, Earth: $M = 5.98 \times 10^{24}$ kg, $r = 6.38 \times 10^6$ m; Sun: $M = 1.99 \times 10^{30}$ kg; 1 AU = 1.50×10^{11} m; 1 year = 365.25 days; $g = 9.81$ m/s^2.

• **Signs:** forces are written as sizes, + toward the centre. Every step keeps full precision; only final answers are rounded.

QUESTION 1

Mass of Jupiter from its moon Io

1. An astronomer observes the planet Jupiter and finds that the period of its moon Io is 1.736 days. If this moon has an average radius of orbit about Jupiter of 4.2 x 10^8 m, what is the mass of Jupiter, using these measurements? [1.95 x 10^{27} kg]

Step 1 – Period in seconds

$$T = (1.736\ \text{d})(24\ \text{h/d})(3600\ \text{s/h}) = 1.4999 \times 10^5\ \text{s}$$

Step 2 – Gravity is the centripetal force

Use the period form because T is known. Io's mass m cancels:

$$\frac{GMm}{r^2} = \frac{m4\pi^2 r}{T^2}$$

Divide by m, multiply by r^2 and divide by G: $M = \dfrac{4\pi^2 r^3}{GT^2}$

$$M = \frac{4\pi^2(4.2\times10^8)^3}{(6.67\times10^{-11})(1.4999\times10^5)^2} = 1.949\times10^{27}\ \text{kg}$$

FINAL ANSWER

$M_{Jupiter} = 1.9 \times 10^{27}$ kg

Worksheet answer: 1.95×10^{27} kg
Why the answers differ: The physics is the same; only the rounding differs. The unrounded mass is 1.949×10^{27} kg, which is the worksheet's value to 3 figures. The radius 4.2×10^8 m has only 2 significant figures, so the answer is 1.9×10^{27} kg.

QUESTION 2

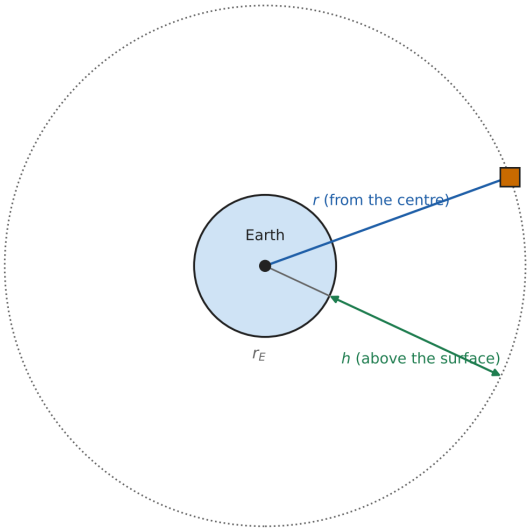
Geosynchronous satellite

2. Communications satellites are placed in orbit so that they remain stationary relative to a specific area on the Earth's surface. They are given the name *geosynchronous satellites* because, to maintain such a position, their period as they orbit must be the same as Earth's. What is the height of such a satellite measured from the following:

a. the centre of Earth? [4.22 x 10^7 m from centre of earth]

b. the surface of Earth?

Picture the situation



The orbit radius r is measured from Earth's centre; the height h is measured from the surface.

a. the centre of Earth?

The period is one day, T = 24 h = 86 400 s. Set gravity equal to the centripetal force and solve for r:

$$\frac{GMm}{r^2} = \frac{m4\pi^2 r}{T^2} \quad \Rightarrow \quad r^3 = \frac{GMT^2}{4\pi^2}$$

$$r^3 = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(86\,400)^2}{4\pi^2} = 7.542 \times 10^{22}\ \text{m}^3$$

Cube root both sides: $r = 4.225 \times 10^7$ m

b. the surface of Earth?

Subtract Earth's radius:

$$h = r - r_E = 4.225 \times 10^7 - 6.38 \times 10^6 = 3.587 \times 10^7\ \text{m}$$

FINAL ANSWER

a. r = 4.23 × 10^7 m from the centre of Earth

b. h = 3.59 × 10^7 m above the surface

Worksheet answer: a. 4.22 × 10^7 m
Why the answers differ: Here the worksheet is **not** wrong; it is slightly more accurate than this key. Earth turns once relative to the stars every 23 h 56 min (86 164 s), a little less than the 24 h day used above. With T = 86 164 s, r = 4.217 × 10^7 m = 4.22 × 10^7 m (and h = 3.58 × 10^7 m). With T = 24 h, as in this key, r = 4.225 × 10^7 m = 4.23 × 10^7 m. The two differ by less than 0.2%, and both round to 4.2 × 10^7 m.

QUESTION 3

Stars orbiting a galaxy

3. Astronomers observing the light from a galaxy calculate the mass of the galaxy to be 2.55 x 10^{41} kg within a radius of 3.24 x 10^{20} m.

a. With this mass, how fast would they expect the stars orbiting the galaxy at this radius to be moving? [2.29 x 10^{5} m/s]

b. If the visible mass only represents 10% of the actual mass of the galaxy, how fast would they actually observe the orbiting stars to be moving? [7.24 x 10^{5} m/s]

c. Why is your answer for (b) not simply 10x your answer for (a)?

a. how fast would they expect the stars orbiting the galaxy at this radius to be moving?

Gravity from the galaxy's mass is the centripetal force on a star; the star's mass cancels:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{\frac{GM}{r}}$$

$$v = \sqrt{\frac{(6.67 \times 10^{-11})(2.55 \times 10^{41})}{3.24 \times 10^{20}}} = 2.291 \times 10^{5} \text{ m/s}$$

b. how fast would they actually observe the orbiting stars to be moving?

If the visible mass is 10% of the total, the total mass is 10 times bigger:

$$M = 10(2.55 \times 10^{41}) = 2.55 \times 10^{42} \text{ kg}$$

$$v = \sqrt{\frac{(6.67 \times 10^{-11})(2.55 \times 10^{42})}{3.24 \times 10^{20}}} = 7.245 \times 10^{5} \text{ m/s}$$

c. Why is your answer for (b) not simply 10x your answer for (a)?

The speed depends on the square root of the mass, v = √(GM/r). Ten times the mass gives √10 ≈ 3.16 times the speed:

$$\frac{v_b}{v_a} = \sqrt{10} = 3.162 \qquad (2.291 \times 10^5)(3.162) = 7.245 \times 10^5$$

FINAL ANSWER

a. v = 2.29 × 10^5 m/s

b. v = 7.25 × 10^5 m/s

c. v ∝ √M, so 10 times the mass gives only √10 ≈ 3.16 times the speed

Worksheet answer: b. 7.24 × 10^5 m/s
Why the answers differ: The worksheet rounded twice. Its value is what you get by multiplying the **rounded** answer to (a) by √10: (2.29 × 10^5)(3.162) = 7.24 × 10^5. Using the unrounded (a), 2.2912 × 10^5 m/s, gives 7.2454 × 10^5 m/s, which rounds to 7.25 × 10^5 m/s. Rounding an answer and then using it in the next step adds error.

QUESTION 4

Venus’s distance from the Sun

4. The planet Venus has a nearly circular orbit. We can observe from Earth that one Venusian “year” is equal to about 225 earth days.

a. If Venus’ mass is 4.87 x 10^{24} kg, determine its distance from the Sun in Astronomical Units (AU – the distance from the earth to the Sun) (0.7 AU)

b. Derive an expression for the radius of any satellite’s orbit around another object in terms of its period, the gravitational constant and the mass of the body being orbited. [1.08 x 10^{11} m]

b. Derive an expression for the radius of any satellite’s orbit (done first, since (a) uses it)

Gravity from the central body (mass M) is the centripetal force on the satellite (mass m):

$$\frac{GMm}{r^2} = \frac{m4\pi^2 r}{T^2}$$

Divide both sides by m: $\frac{GM}{r^2} = \frac{4\pi^2 r}{T^2}$

Multiply both sides by r^2T^2 and divide by $4\pi^2$: $r^3 = \frac{GMT^2}{4\pi^2}$

Cube root both sides: $r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$

The satellite’s own mass cancels, so Venus’s mass (4.87×10^{24} kg) isn’t needed. M is the Sun’s mass.

a. determine its distance from the Sun in Astronomical Units

T = 225 days = 1.944×10^{7} s. Use the formula from (b) with the Sun's mass:

$$r = \sqrt[3]{\frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(1.944 \times 10^{7})^{2}}{4\pi^{2}}} = 1.083 \times 10^{11} \text{ m}$$

$$\text{In AU:} \quad \frac{1.083 \times 10^{11} \text{ m}}{1.50 \times 10^{11} \text{ m/AU}} = 0.722 \text{ AU}$$

FINAL ANSWER

a. r = 1.08 × 10^{11} m = 0.72 AU (about 0.7 AU)

b. r = $\sqrt[3]{(GMT^{2} \div 4\pi^{2})}$, where M is the mass of the body being orbited

Worksheet answer: a. 0.7 AU; b. [1.08 × 10^{11} m]
Why the answers differ: (a) Same value: 1.083 × 10^{11} m ÷ 1.50 × 10^{11} m = 0.722 AU. The worksheet rounds to 1 figure; the key gives 2 (0.72 AU). (b) asks for an expression, not a number. The worksheet's [1.08 × 10^{11} m] is the answer to (a) in metres, printed under the wrong part. Venus's mass isn't needed at all, because it cancels.

QUESTION 5

Lunar lander in orbit

5. A lunar lander is to be placed in orbit around the moon at a mean altitude of 100 km. What will be the period of the lunar lander? (m_{moon} = 7.34 x 10^{22} kg; r_{moon} = 1.74 x 10^{6} m) [1.97 h]

The orbit radius is measured from the Moon's centre: r = 1.74 × 10^{6} m + 1.00 × 10^{5} m = 1.84 × 10^{6} m. Rearrange $r^3 = GMT^2/4\pi^2$ for T:

$$T^2 = \frac{4\pi^2 r^3}{GM} \quad \Rightarrow \quad T = 2\pi\sqrt{\frac{r^3}{GM}}$$

$$T = 2\pi\sqrt{\frac{(1.84\times10^{6})^3}{(6.67\times10^{-11})(7.34\times10^{22})}} = 7088\ \text{s}$$

In hours: $\dfrac{7088\ \text{s}}{3600\ \text{s/h}} = 1.969\ \text{h}$

FINAL ANSWER

T = 1.97 h

QUESTION 6

Satellite with a 36 hour orbit

6. A satellite orbits Earth once every 36 hours. Calculate its velocity and acceleration. [2.68 x 10^{3} m/s; 0.13 m/s^2]

Step 1 – Orbit radius

T = 36 h = 129 600 s. As in Question 2:

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}} = \sqrt[3]{\frac{(6.67\times10^{-11})(5.98\times10^{24})(129\,600)^2}{4\pi^2}} = 5.536\times10^{7}\ \text{m}$$

Step 2 – Speed and acceleration

$$v = \frac{2\pi r}{T} = \frac{2\pi(5.536 \times 10^7)}{129\,600} = 2.684 \times 10^3 \text{ m/s}$$

$$a_c = \frac{v^2}{r} = \frac{(2.684 \times 10^3)^2}{5.536 \times 10^7} = 0.1301 \text{ m/s}^2$$

FINAL ANSWER

v = 2.7 × 10^3 m/s (along the orbit)

a = 0.13 m/s^2 toward Earth's centre

Worksheet answer: 2.68 × 10^3 m/s
Why the answers differ: Same value, different rounding. The unrounded speed is 2684 m/s. "36 hours" has 2 significant figures, so the key gives 2.7 × 10^3 m/s; the worksheet keeps 3 figures (while giving the acceleration, 0.13 m/s^2, to 2).

QUESTION 7

Uranus's orbit

7. The orbit of Uranus around the Sun is nearly a circle of radius 2.87 x 10^{12} m. The speed of Uranus is 6.80 x 10^3 m/s and its mass is 8.80 x 10^{25} kg.

a) Name the force that causes the centripetal acceleration.

b) Determine the magnitude of this force. [1.42 x 10^{21} N]

c) Calculate the orbital period of Uranus in seconds and in earth years. [2.65 x 10^9 s; 84 y]

a)

The gravitational force of the Sun on Uranus.

b)

$$F_c = \frac{mv^2}{r} = \frac{(8.80 \times 10^{25})(6.80 \times 10^3)^2}{2.87 \times 10^{12}} = 1.418 \times 10^{21} \text{ N}$$

c)

One orbit is the circumference, 2πr, at speed v:

$$T = \frac{2\pi r}{v} = \frac{2\pi(2.87 \times 10^{12})}{6.80 \times 10^{3}} = 2.652 \times 10^{9}\ \text{s}$$

In years: $\dfrac{2.652 \times 10^{9}\ \text{s}}{(365.25)(86\,400)\ \text{s/y}} = 84.0\ \text{y}$

FINAL ANSWER

a) Gravity from the Sun

b) F = 1.42 × 10^{21} N

c) T = 2.65 × 10^{9} s = 84 earth years

QUESTION 8

Roller coaster loop

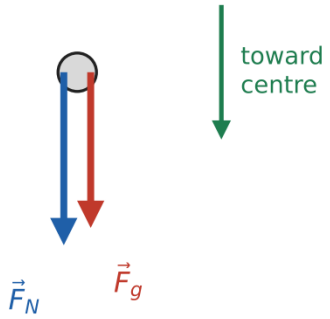
8. A roller coaster has a loop of radius 12 m.

   a) Determine the speed required by the coaster that would cause a rider to feel twice their normal weight at the top of the loop. [19 m/s]

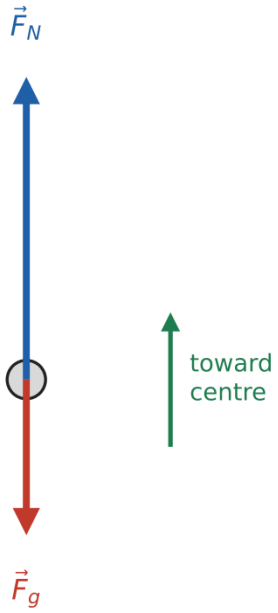
   b) How many times their normal weight would the same rider feel at the bottom of the loop? [4x]

FBDs of the rider (the seat pushes toward the centre)

a. top of the loop



b. bottom of the loop



a) the speed that makes a rider feel twice their normal weight at the top

Feeling twice as heavy means $F_N = 2mg$. At the top, F_N and gravity both point down, toward the centre:

$$F_N + mg = \frac{mv^2}{r} \quad \Rightarrow \quad 2mg + mg = \frac{mv^2}{r}$$

Divide by m and multiply by r: $\quad v^2 = 3gr$

$$v = \sqrt{3(9.81)(12)} = 18.79 \text{ m/s}$$

b) How many times their normal weight would the same rider feel at the bottom of the loop?

Take the same speed at the bottom (as the worksheet's answer does), so $mv^2/r = 3mg$ again. At the bottom, F_N points up (toward the centre) and gravity points down:

$$F_N - mg = \frac{mv^2}{r} = 3mg \quad \Rightarrow \quad F_N = 4mg$$

(A real coaster is faster at the bottom, so the rider would feel even heavier.)

FINAL ANSWER

a) v = 19 m/s

b) 4 times their normal weight

QUESTION 9

Bird pulling out of a dive

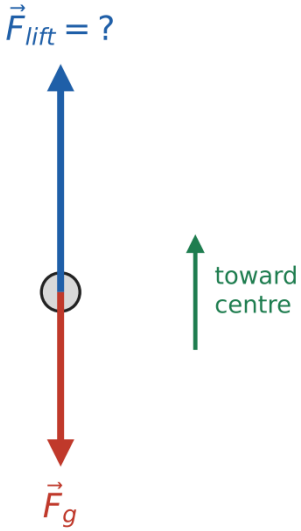
9. A bird of mass 0.211 kg pulls out of a dive, the bottom of which can be considered to be a circular arc of radius 25.6 m. At the bottom of the arc, the bird's speed is a constant 21.7 m/s. Determine the magnitude of the upwards lift on the bird's wings at the bottom of the arc. [5.95 N]

At the bottom of the arc the centre is above the bird. Lift points up (toward the centre) and gravity down:

$$F_{lift} - mg = \frac{mv^2}{r} \quad \Rightarrow \quad F_{lift} = m\left(g + \frac{v^2}{r}\right)$$

$$F_{lift} = (0.211)\left(9.81 + \frac{(21.7)^2}{25.6}\right) = (0.211)(9.81 + 18.394) = 5.951 \text{ N}$$

FBD of the bird at the bottom of the arc



FINAL ANSWER

F_{lift} = 5.95 N

QUESTION 10

Banking angle for a slippery curve

10. A highway curve in the horizontal plane is banked so that vehicles can proceed safely even if the road is slippery. Determine the proper banking angle for a car travelling at 97 km/h on a curve of radius 450 m. [9.3°]

No friction (slippery), so use $\tan\theta = v^2/(rg)$ from 2.8, Example 2. 97 km/h ÷ 3.6 = 26.94 m/s:

$$\tan\theta = \frac{(26.944)^2}{(450)(9.81)} = 0.1645 \quad \Rightarrow \quad \theta = 9.34°$$

FINAL ANSWER

θ = 9.3°

QUESTION 11

Stone on a rope on a frictionless surface

11. A 2.00 kg stone attached to a rope 4.00 m long is whirled in a circle horizontally on a frictionless surface, completing 5.00 revolutions in 2.00 s. Determine the tension in the rope. [1.97 x 10^3 N]

f = 5.00 rev ÷ 2.00 s = 2.50 Hz. The rope is horizontal, so the tension is the centripetal force:

$$T = m(4\pi^2 rf^2) = (2.00)(4\pi^2)(4.00)(2.50)^2 = 1973.9 \text{ N}$$

FINAL ANSWER

T = 1.97 × 10^3 N

QUESTION 12

Plane at the top of a vertical loop

12. A plane is flying in a vertical loop of radius 1.50 km. At what speed is the plane flying at the top of the loop if the vertical force exerted by the air on the plane is zero at this point? State your answer both in m/s and in km/h [121 m/s; 437 km/h]

With no force from the air, gravity alone is the centripetal force (r = 1.50 km = 1500 m):

$$mg = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{gr} = \sqrt{(9.81)(1500)} = 121.31 \text{ m/s}$$

In km/h: $(121.31)(3.6) = 436.7$ km/h

FINAL ANSWER

v = 121 m/s = 437 km/h